

1. Questions

Study the following information carefully and answer the following questions.

The table below shows the total number of currency notes, the ratio of the number of the Rs. 'X' notes to the Rs. 'Y' notes, the sum of the number of Rs. Y notes and the number of Rs. Z notes and the total amount deposited in a bank by the four different persons.

Persons	The total number of currency notes	The ratio of the Rs. 'X' notes to the Rs. 'Y' notes	The sum of the number of Rs. Y notes and the number of Rs. Z notes	The total amount deposited (Rs.)
Raj	12	-	7	6900
Kavin	-	3: 4	24	19440
Suriya	27	3: 4	-	-
Veera	-	1: 2	35	34200

Notes:

i). The denomination of Rs. 'Y' is 10 times that of the Rs. 'X' notes, and the denomination of Rs. 'Z' is 10 times that of the Rs. 'Y' notes.

ii). The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 5, 10, 20, 50, 100, 200, 500, 1000, and 2000.

Which of the following statements is definitely not true?

I: The total number of notes deposited by Kavin is a perfect square.

II: The total number of notes deposited by all persons is multiple of 3.

III: The sum of the number of Rs. X notes deposited by Suriya and Veera together is 19.

IV: The total amount deposited by Suriya is 25% of that by Kavin.

- Only I
- Only III
- Only II
- Only IV
- Both (b) and (d)

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2. Questions

If the number of Rs. Z notes deposited in a bank by Suriya is 86.66% of the number of Rs. Z notes deposited by Veera. Find the difference between the total amount deposited by Suriya and that deposited by Kavin.

- Rs. 8280

- b. Rs. 8880
- c. Rs. 8180
- d. Rs. 7280
- e. Rs. 6280

3. Questions

If the amount deposited by Raj in a bank is increased to Rs. $(4Z + 2Y)$ with the addition of the three new denominations of Rs. M notes. Which of the following statements is true about the value of M ? (M is a single digit number)

I: $M < X$

II: $M > Z$

III: $M > Y$

- a. Both I and II
- b. Both II and III
- c. Only III
- d. Only I
- e. Only II

4. Questions

If the bank assigns an interest rate of $(X - 5)\%$ simple interest per annum to Veera. The interest amount received by Veera after 3 years is what percentage of the $10Z$?

- a. 75%
- b. 72.5%
- c. 77.77%
- d. 76.95%
- e. None of these

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5. Questions

If the number of Rs. X notes deposited by Mani is 25% more than that of Kavin, the number of Rs. Y notes deposited by Mani is 50% more than that of Raj and he has deposited some Rs. Z notes. If the total amount deposited by Mani is Rs. 25500. Which of the following persons has the second highest total number of notes deposited in a bank?

- a. Raj
- b. Kavin
- c. Suriya

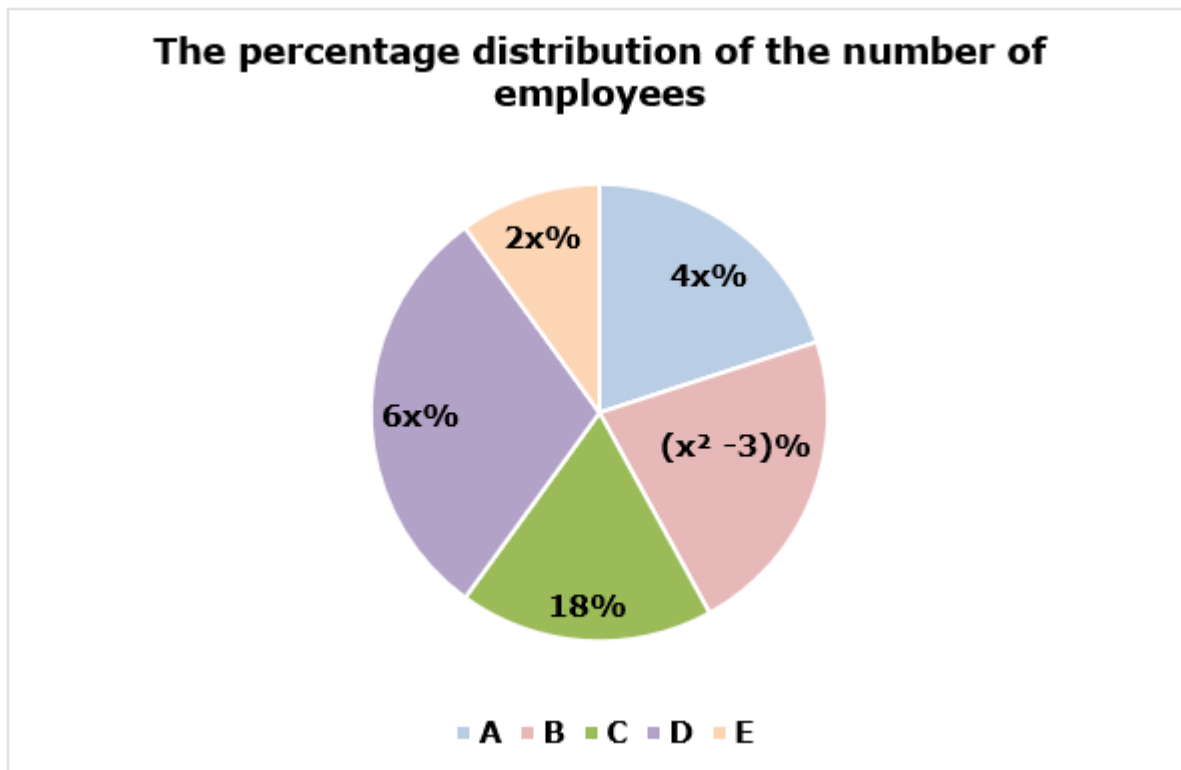
d. Veera

e. Mani

6. Questions

Study the following information carefully and answer the following questions.

The pie chart below shows the percentage distribution of the number of employees in each company out of the total number of employees in five different companies combined.



The table below shows the percentage of full-time employees in each company and the ratio of full-time male employees to intern female employees in five different companies.

Company	The percentage of full-time employees	The ratio of full-time male employees to intern female employees
A	60%	5:1
B	72.72%	85: 18
C	66.66%	4: 1
D	40%	12: 17
E	80%	17: 2

Notes:

i). The total number of (male + female) employees = The number of (male + female) full-time employees + the number of (male + female) intern employees

ii). The total number of employees in five companies together is 'N * 10x'

iii). The value of 'N' is a third smallest two-digit prime number.

If the number of full-time female employees and the number of intern female employees are same in company A. If the number of intern male employees in company A is same as the number of full-time female employees in company B. Find the average number of intern male employees in companies A and B together.

- a. 42
- b. 38
- c. 55
- d. 47
- e. 53

7. Questions

If the total number of intern employees in company B is equal to the total number of full-time female employees in the same company. Which of the following statements is true?

I: The number of intern male employees in company B is a multiple of 3

II: The number of intern female employees in company B is multiple of 2

III: The number of full-time male employees in company B is multiple of 5

- a. Both I and II
- b. Both II and III
- c. All I, II and III
- d. Both I and III
- e. None is true

8. Questions

The sum of the full-time employees in companies D and E together is what percentage of the sum of the intern employees in companies A, B and C together?

- a. 20x%
- b. 100%
- c. 75%
- d. Both (a) and (c)
- e. Both (a) and (b)

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9. Questions

If the sum of the number of full-time male employees in companies C and D together is $28x$ and the sum of the number of intern female employees in companies C and D together ' $20x + 19$ '. Which of

the following statements is true?

I: The number of full-time male employees in company C > The number of full-time male employees in company D

II: The number of full-time female employees in company C > The number of intern male employees in company D

III: The number of intern female employees in company C < The number of full-time female employees in company D

- a. Both I and II
- b. Both II and III
- c. All I, II and III
- d. Only III
- e. Only II

10. Questions

If the total number of employees in company F is $4x\%$ more than that of company E. The ratio of number of full-time employees to the number of intern employees in company F is 12: 5, respectively. If the number of full-time female employees in company E is 4 more than the number of intern employees in company F, what is the ratio of the number full-time female employees to the number of intern male employees in company E?

- a. 12: 5
- b. 1: 1
- c. 34: 13
- d. 17: 2
- e. Cannot be determined.

11. Questions

Study the following information carefully and answer the following questions.

The information below shows that the shopkeeper sells three different products viz., A, B and C to the customer.

The cost price of product A is $\frac{A}{100}\%$ less than the cost price of product B which marked up 54% above its cost price. The ratio of the discount percentage of products A and C is $\frac{B}{D}\%$ respectively and the ratio of the discount percentage of products B and C is 4: 5, respectively. The marked price of product C is $(3x + 5)\%$ more than its cost price and the discount percentage of the product C is $x\%$. The profit amount of product C is Rs. $\frac{C^2}{100}$. The profit percentage of product A is 37.7% and the discount percentage of the product A is $(x - 5)\%$. The difference between the cost price of products A and C is Rs. $\frac{E}{100}$. The marked up price of the product A is Rs. $62x$.

Notes:

- i). The ratio of the selling price of products B and C is 49:36, respectively.
- ii). The value of 'x' is a two-digit number and 'x' is a multiple of 10.
- iii). $10 < x < 40$
- iv). The cost price of the product C is Rs. 90x
- v) The cost price, marked price and selling price of the each article are integers.

Which of the following statements is true?

I: The selling price of the product B is a multiple of D

II: The value of $(A + B)$ is a prime number.

III: The value of 'x' is a multiple of B.

- a. Only II
- b. Only I
- c. Only III
- d. Both I and II
- e. Both I and III

12. Questions

If the shopkeeper added the additional discount of $(A + 5)\%$ on product C. Find the new profit or loss percentage of the product C.

- a. $(B - 2)\%$
- b. 1%
- c. $(D - 2)\%$
- d. Both (b) and (c)
- e. Both (a) and (b)

13. Questions

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If the difference between the selling price of the products A and B is Rs. F. Find the wrong number of the below series.

Series: D, A, C, $(E - 75)$, 130, $(F + 170)$

- a. D
- b. A
- c. $(F + 170)$
- d. C

e. $(E - 75)$

14. Questions

If the cost price of the product E is $x\%$ more than the cost price of the product C. The marked up percentage of the product E is 50% and gives a two successive discount of $A\%$ and $(C + 1)\%$. Find the profit or loss percentage of the product E.

- a. $A\%$
- b. $D\%$
- c. 10%
- d. 8%
- e. None of these

15. Questions

If $C\%$ of the average of the marked prices of products A, B, and C together is the selling price of product D. If product D is sold at a loss of 20% , find the cost price of product C.

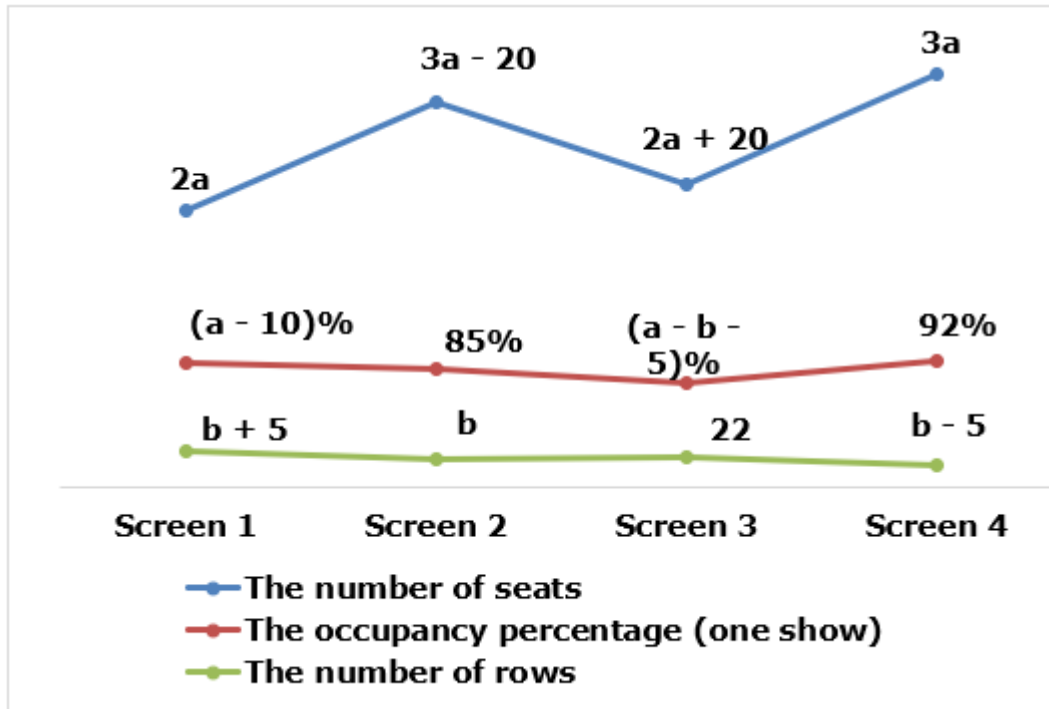
- a. Rs. 1000
- b. Rs. 2006
- c. Rs. 1506
- d. Rs. 1408
- e. Rs. 1006

16. Questions

Study the following information carefully and answer the following questions.

The line graph below shows the number of seats, the occupancy percentage (one show), and the number of rows in four screens in a multiplex.

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Notes:

- The occupancy percentage = (the number of tickets booked / the total number of seats) * 100
- The total number of seats in each screen = the number of rows * the number of seats in each row.
- The number of seats in each row in screen 1 is 8 and the number of seats in each row in screen 4 is 20.

If the 20% of the rows in a screen 1 is Executive seats rows, others are normal seats row. If the ticket price of executive seat and normal seat in screen 1 is Rs. 350 and Rs. 200 respectively. Find the total revenue generated from screen 1 in a particular show.

Note: The total number of not booked tickets is normal seats in screen 1 in a particular show.

- Rs. 46000
- Rs. 42000
- Rs. 48000
- Rs. 40000
- Rs. 52000

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17. Questions

If the percentage of the total number of seats booked in screen 2 by online is 57.14% of the total number of seats booked in screen 2 and other seats booked in screen 2 by offline. Find the number of seats booked through offline is what percentage of the total number of seats in Screen 4?

- $(a - 66)\%$
- 34% of a
- $(b + 14)\%$

- d. Both (b) and (c)
- e. All (a), (b) and (c)

18. Questions

Which of the following statements is true?

I: The number of rows in screen 3 is 22% of the value of 'a'.

II: The number of not booked seats in screen 1 is 'b'.

III: The number of booked seats in screen 4 is ' $2a + 76$ '.

- a. Both I and II
- b. All I, II and III
- c. Both II and III
- d. Only I
- e. Only II

19. Questions

If the total number of seats in screen 5 is $(b + 5)\%$ less than the total number of seats in screen 4.

If the number of rows and the number of seats in each row in screen 5 are equal. Find the total number of rows in screens 3, 4, and 5 together.

- a. 58
- b. 44
- c. 52
- d. 42
- e. 68

20. Questions

What is the ratio of the average number of not booked seats in screens 2 and 4 together in a particular show to the total number of seats in each row in screens 1, 2, 3, and 4 together?

- a. 22: 51
- b. 33: 56
- c. 11: 21
- d. 25: 52
- e. 33: 52

Explanations:

1. Questions

The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' notes be (5, 50, 500) or (10, 100, 1000) or (20, 200, 2000) respectively.

We take from Raj's deposit amount

The total number of currency notes = 12

The sum of the number of Rs. Y notes and the number of Rs. Z notes = 7

The number of Rs. X notes = $12 - 7 = 5$

The number of Rs. Y notes = a

The number of Rs. Z notes = $7 - a$

Case I: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 5, 50, 500)

$$5 * 5 + 50 * a + 500 * (7 - a) = 6900$$

$$25 + 50a + 3500 - 500a = 6900$$

$$3525 - 450a = 6900$$

$$-450a = 3375$$

This is not satisfied because it is negative value.

Case II: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 10, 100, 1000)

$$10 * 5 + 100 * a + 1000 * (7 - a) = 6900$$

$$50 + 100a + 7000 - 1000a = 6900$$

$$7050 - 900a = 6900$$

$$900a = 150$$

$$a = 150/900$$

This is not satisfied because it is not an integer.

Case III: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 20, 200, 2000)

$$20 * 5 + 200 * a + 2000 * (7 - a) = 6900$$

$$100 + 200a + 14000 - 2000a = 6900$$

$$14100 - 1800a = 6900$$

$$1800a = 7200$$

$$a = 4$$

This is satisfied.

So, the denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 20, 200, 2000 respectively

$X = 20$, $Y = 200$ and $Z = 2000$

Raj:

The number of Rs. 20 notes = $12 - 7 = 5$

The number of Rs. 200 notes = 4

The number of Rs. 2000 notes = $7 - 4 = 3$

The total amount deposited in a bank = $5 * 20 + 4 * 200 + 3 * 2000 = \text{Rs. } 6900$

Kavin:

The sum of the number of Rs. 200 notes and the number of Rs. 2000 notes = 24

The ratio of the Rs. 20 notes to the Rs. 200 notes = $3:4 = 3b:4b$

The number of Rs. 2000 notes = $24 - 4b$

Let,

$$20 * 3b + 4b * 200 + (24 - 4b) * 2000 = 19440$$

$$60b + 800b + 48000 - 8000b = 19440$$

$$7140b = 28560$$

$$b = 4$$

The number of Rs. 20 notes = $3b = 12$

The number of Rs. 200 notes = $4b = 16$

The number of Rs. 2000 notes = $24 - 16 = 8$

Suriya:

The total number of currency notes = 27

The ratio of the Rs. 20 notes to the Rs. 200 notes = $3:4 = 3c:4c$

The number of Rs. 2000 notes = $27 - 3c - 4c = 27 - 7c$

We cannot find the further more data

Veera:

The sum of the number of Rs. 200 notes and the number of Rs. 2000 notes = 35

The ratio of the Rs. 20 notes to the Rs. 200 notes = $1:2 = d:2d$

The number of Rs. 2000 notes = $35 - 2d$

Let

$$20 * d + 2d * 200 + (35 - 2d) * 2000 = 34200$$

$$20d + 400d + 70000 - 4000d = 34200$$

$$3580d = 35800$$

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$d = 10$

The number of Rs. 20 notes = $d = 10$

The number of Rs. 200 notes = $2d = 20$

The number of Rs. 2000 notes = $35 - 20 = 15$

Persons	The total number of notes	The number of Rs. 20 notes	The number of Rs. 200 notes	The number of Rs. 2000 notes	The total amount deposited (Rs.)
Raj	12	5	4	3	6900
Kavin	36	12	16	8	19440
Suriya	27	-	-	-	-
Veera	45	10	20	15	34200

Answer: E

Let according to the question

I: The total number of currency notes deposited by Kavin is a perfect square.

36 is a perfect square. This is true.

II: The total number of notes deposited by all persons is multiple of 3.

12, 36, 27, and 45 are multiple of 3 and their sum 120 is also a multiple of 3. This is true.

III: The sum of the number of Rs. X notes deposited by Suriya and Veera together is 19.

We do not know the number of Rs. 20 notes deposited by Suriya. This is not true.

IV: The total amount deposited by Suriya is 25% of that by Kavin.

We do not know the total amount deposited by Suriya. This is not true.

2. Questions

The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' notes be (5, 50, 500) or (10, 100, 1000) or (20, 200, 2000) respectively.

We take from Raj's deposit amount

The total number of currency notes = 12

The sum of the number of Rs. Y notes and the number of Rs. Z notes = 7

The number of Rs. X notes = $12 - 7 = 5$

The number of Rs. Y notes = a

The number of Rs. Z notes = $7 - a$

Case I: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 5, 50, 500)

$$5 * 5 + 50 * a + 500 * (7 - a) = 6900$$

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$$25 + 50a + 3500 - 500a = 6900$$

$$3525 - 450a = 6900$$

$$-450a = 3375$$

This is not satisfied because it is negative value.

Case II: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 10, 100, 1000)

$$10 * 5 + 100 * a + 1000 * (7 - a) = 6900$$

$$50 + 100a + 7000 - 1000a = 6900$$

$$7050 - 900a = 6900$$

$$900a = 150$$

$$a = 150/900$$

This is not satisfied because it is not an integer.

Case III: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 20, 200, 2000)

$$20 * 5 + 200 * a + 2000 * (7 - a) = 6900$$

$$100 + 200a + 14000 - 2000a = 6900$$

$$14100 - 1800a = 6900$$

$$1800a = 7200$$

$$a = 4$$

This is satisfied.

So, the denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 20, 200, 2000 respectively

$$X = 20, Y = 200 \text{ and } Z = 2000$$

Raj:

$$\text{The number of Rs. 20 notes} = 12 - 7 = 5$$

$$\text{The number of Rs. 200 notes} = 4$$

$$\text{The number of Rs. 2000 notes} = 7 - 4 = 3$$

$$\text{The total amount deposited in a bank} = 5 * 20 + 4 * 200 + 3 * 2000 = \text{Rs. 6900}$$

Kavin:

$$\text{The sum of the number of Rs. 200 notes and the number of Rs. 2000 notes} = 24$$

$$\text{The ratio of the Rs. 20 notes to the Rs. 200 notes} = 3: 4 = 3b: 4b$$

$$\text{The number of Rs. 2000 notes} = 24 - 4b$$

Let,

$$20 * 3b + 4b * 200 + (24 - 4b) * 2000 = 19440$$

$$60b + 800b + 48000 - 8000b = 19440$$

$$7140b = 28560$$

$$b = 4$$

$$\text{The number of Rs. 20 notes} = 3b = 12$$

$$\text{The number of Rs. 200 notes} = 4b = 16$$

$$\text{The number of Rs. 2000 notes} = 24 - 16 = 8$$

Suriya:

$$\text{The total number of currency notes} = 27$$

$$\text{The ratio of the Rs. 20 notes to the Rs. 200 notes} = 3: 4 = 3c: 4c$$

$$\text{The number of Rs. 2000 notes} = 27 - 3c - 4c = 27 - 7c$$

We cannot find the further more data

Veera:

$$\text{The sum of the number of Rs. 200 notes and the number of Rs. 2000 notes} = 35$$

$$\text{The ratio of the Rs. 20 notes to the Rs. 200 notes} = 1: 2 = d: 2d$$

$$\text{The number of Rs. 2000 notes} = 35 - 2d$$

Let

$$20 * d + 2d * 200 + (35 - 2d) * 2000 = 34200$$

$$20d + 400d + 70000 - 4000d = 34200$$

$$3580d = 35800$$

$$d = 10$$

$$\text{The number of Rs. 20 notes} = d = 10$$

$$\text{The number of Rs. 200 notes} = 2d = 20$$

$$\text{The number of Rs. 2000 notes} = 35 - 20 = 15$$

Persons	The total number of notes	The number of Rs. 20 notes	The number of Rs. 200 notes	The number of Rs. 2000 notes	The total amount deposited (Rs.)
Raj	12	5	4	3	6900
Kavin	36	12	16	8	19440
Suriya	27	-	-	-	-
Veera	45	10	20	15	34200

Answer: A

The number of Rs. 2000 notes deposited by Suriya = 86.66% of 15 = $13/15 * 15 = 13$

The sum of Rs. 20 notes and Rs. 200 notes deposited = $27 - 13 = 14$

The number of Rs. 20 notes deposited = $14/7 * 3 = 6$

The number of Rs. 200 notes deposited = $14/7 * 4 = 8$

The total amount deposited by Suriya = $6 * 20 + 8 * 200 + 13 * 2000 = \text{Rs. } 27720$

Thus, difference = $27720 - 19440 = \text{Rs. } 8280$

3. Questions

The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' notes be (5, 50, 500) or (10, 100, 1000) or (20, 200, 2000) respectively.

We take from Raj's deposit amount

The total number of currency notes = 12

The sum of the number of Rs. Y notes and the number of Rs. Z notes = 7

The number of Rs. X notes = $12 - 7 = 5$

The number of Rs. Y notes = a

The number of Rs. Z notes = $7 - a$

Case I: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 5, 50, 500)

$$5 * 5 + 50 * a + 500 * (7 - a) = 6900$$

$$25 + 50a + 3500 - 500a = 6900$$

$$3525 - 450a = 6900$$

$$-450a = 3375$$

This is not satisfied because it is negative value.

Case II: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 10, 100, 1000)

$$10 * 5 + 100 * a + 1000 * (7 - a) = 6900$$

$$50 + 100a + 7000 - 1000a = 6900$$

$$7050 - 900a = 6900$$

$$900a = 150$$

$$a = 150/900$$

This is not satisfied because it is not an integer.

Case III: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 20, 200, 2000)

$$20 * 5 + 200 * a + 2000 * (7 - a) = 6900$$

$$100 + 200a + 14000 - 2000a = 6900$$

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$$14100 - 1800a = 6900$$

$$1800a = 7200$$

$$a = 4$$

This is satisfied.

So, the denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 20, 200, 2000 respectively

$$X = 20, Y = 200 \text{ and } Z = 2000$$

Raj:

$$\text{The number of Rs. 20 notes} = 12 - 7 = 5$$

$$\text{The number of Rs. 200 notes} = 4$$

$$\text{The number of Rs. 2000 notes} = 7 - 4 = 3$$

$$\text{The total amount deposited in a bank} = 5 * 20 + 4 * 200 + 3 * 2000 = \text{Rs. } 6900$$

Kavin:

$$\text{The sum of the number of Rs. 200 notes and the number of Rs. 2000 notes} = 24$$

$$\text{The ratio of the Rs. 20 notes to the Rs. 200 notes} = 3: 4 = 3b: 4b$$

$$\text{The number of Rs. 2000 notes} = 24 - 4b$$

Let,

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$$60b + 800b + 48000 - 8000b = 19440$$

$$7140b = 28560$$

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$$\text{The number of Rs. 20 notes} = 3b = 12$$

$$\text{The number of Rs. 200 notes} = 4b = 16$$

$$\text{The number of Rs. 2000 notes} = 24 - 16 = 8$$

Suriya:

$$\text{The total number of currency notes} = 27$$

$$\text{The ratio of the Rs. 20 notes to the Rs. 200 notes} = 3: 4 = 3c: 4c$$

$$\text{The number of Rs. 2000 notes} = 27 - 3c - 4c = 27 - 7c$$

We cannot find the further more data

Veera:

$$\text{The sum of the number of Rs. 200 notes and the number of Rs. 2000 notes} = 35$$

$$\text{The ratio of the Rs. 20 notes to the Rs. 200 notes} = 1: 2 = d: 2d$$

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The number of Rs. 2000 notes = $35 - 2d$

Let

$$20 * d + 2d * 200 + (35 - 2d) * 2000 = 34200$$

$$20d + 400d + 70000 - 4000d = 34200$$

$$3580d = 35800$$

$$d = 10$$

The number of Rs. 20 notes = $d = 10$

The number of Rs. 200 notes = $2d = 20$

The number of Rs. 2000 notes = $35 - 20 = 15$

Persons	The total number of notes	The number of Rs. 20 notes	The number of Rs. 200 notes	The number of Rs. 2000 notes	The total amount deposited (Rs.)
Raj	12	5	4	3	6900
Kavin	36	12	16	8	19440
Suriya	27	-	-	-	-
Veera	45	10	20	15	34200

Answer: C

New deposit amount by Raj = $4 * 2000 + 2 * 200 = \text{Rs. } 8400$

The total amount of denomination of Rs. M notes = $8400 - 6900 = \text{Rs. } 1500$

$$M = 1500/3 = 500$$

$$M = 500$$

I: $M < X = 500 < 20$ (false)

II: $M > Z = 500 > 2000$ (false)

III: $M > Y = 500 > 200$ (true)

4. Questions

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The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' notes be (5, 50, 500) or (10, 100, 1000) or (20, 200, 2000) respectively.

We take from Raj's deposit amount

The total number of currency notes = 12

The sum of the number of Rs. Y notes and the number of Rs. Z notes = 7

The number of Rs. X notes = $12 - 7 = 5$

The number of Rs. Y notes = a

The number of Rs. Z notes = $7 - a$

Case I: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 5, 50, 500)

$$5 * 5 + 50 * a + 500 * (7 - a) = 6900$$

$$25 + 50a + 3500 - 500a = 6900$$

$$3525 - 450a = 6900$$

$$-450a = 3375$$

This is not satisfied because it is negative value.

Case II: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 10, 100, 1000)

$$10 * 5 + 100 * a + 1000 * (7 - a) = 6900$$

$$50 + 100a + 7000 - 1000a = 6900$$

$$7050 - 900a = 6900$$

$$900a = 150$$

$$a = 150/900$$

This is not satisfied because it is not an integer.

Case III: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 20, 200, 2000)

$$20 * 5 + 200 * a + 2000 * (7 - a) = 6900$$

$$100 + 200a + 14000 - 2000a = 6900$$

$$14100 - 1800a = 6900$$

$$1800a = 7200$$

$$a = 4$$

This is satisfied.

So, the denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 20, 200, 2000 respectively

$$X = 20, Y = 200 \text{ and } Z = 2000$$

Raj:

$$\text{The number of Rs. 20 notes} = 12 - 7 = 5$$

$$\text{The number of Rs. 200 notes} = 4$$

$$\text{The number of Rs. 2000 notes} = 7 - 4 = 3$$

$$\text{The total amount deposited in a bank} = 5 * 20 + 4 * 200 + 3 * 2000 = \text{Rs. } 6900$$

Kavin:

$$\text{The sum of the number of Rs. 200 notes and the number of Rs. 2000 notes} = 24$$

$$\text{The ratio of the Rs. 20 notes to the Rs. 200 notes} = 3 : 4 = 3b : 4b$$

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The number of Rs. 2000 notes = $24 - 4b$

Let,

$$20 * 3b + 4b * 200 + (24 - 4b) * 2000 = 19440$$

$$60b + 800b + 48000 - 8000b = 19440$$

$$7140b = 28560$$

$$b = 4$$

The number of Rs. 20 notes = $3b = 12$

The number of Rs. 200 notes = $4b = 16$

The number of Rs. 2000 notes = $24 - 16 = 8$

Suriya:

The total number of currency notes = 27

The ratio of the Rs. 20 notes to the Rs. 200 notes = $3:4 = 3c:4c$

The number of Rs. 2000 notes = $27 - 3c - 4c = 27 - 7c$

We cannot find the further more data

Veera:

The sum of the number of Rs. 200 notes and the number of Rs. 2000 notes = 35

The ratio of the Rs. 20 notes to the Rs. 200 notes = $1:2 = d:2d$

The number of Rs. 2000 notes = $35 - 2d$

Let

$$20 * d + 2d * 200 + (35 - 2d) * 2000 = 34200$$

$$20d + 400d + 70000 - 4000d = 34200$$

$$3580d = 35800$$

$$d = 10$$

The number of Rs. 20 notes = $d = 10$

The number of Rs. 200 notes = $2d = 20$

The number of Rs. 2000 notes = $35 - 20 = 15$

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Persons	The total number of notes	The number of Rs. 20 notes	The number of Rs. 200 notes	The number of Rs. 2000 notes	The total amount deposited (Rs.)
Raj	12	5	4	3	6900
Kavin	36	12	16	8	19440
Suriya	27	-	-	-	-
Veera	45	10	20	15	34200

Answer: D

The interest amount received by Veera after 3 years = $34200 \times 3 \times (20 - 5)/100 = \text{Rs. } 15390$

Thus, percentage = $[15390/(10 \times 2000)] \times 100 = 76.95\%$

5. Questions

The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' notes be (5, 50, 500) or (10, 100, 1000) or (20, 200, 2000) respectively.

We take from Raj's deposit amount

The total number of currency notes = 12

The sum of the number of Rs. Y notes and the number of Rs. Z notes = 7

The number of Rs. X notes = $12 - 7 = 5$

The number of Rs. Y notes = a

The number of Rs. Z notes = $7 - a$

Case I: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 5, 50, 500)

$$5 \times 5 + 50 \times a + 500 \times (7 - a) = 6900$$

$$25 + 50a + 3500 - 500a = 6900$$

$$3525 - 450a = 6900$$

$$-450a = 3375$$

This is not satisfied because it is negative value.

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Case II: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 10, 100, 1000)

$$10 \times 5 + 100 \times a + 1000 \times (7 - a) = 6900$$

$$50 + 100a + 7000 - 1000a = 6900$$

$$7050 - 900a = 6900$$

$$900a = 150$$

$$a = 150/900$$

This is not satisfied because it is not an integer.

Case III: (The possible denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 20, 200, 2000)

$$20 * 5 + 200 * a + 2000 * (7 - a) = 6900$$

$$100 + 200a + 14000 - 2000a = 6900$$

$$14100 - 1800a = 6900$$

$$1800a = 7200$$

$$a = 4$$

This is satisfied.

So, the denominations of Rs. 'X', Rs. 'Y', and Rs. 'Z' are 20, 200, 2000 respectively

$$X = 20, Y = 200 \text{ and } Z = 2000$$

Raj:

$$\text{The number of Rs. 20 notes} = 12 - 7 = 5$$

$$\text{The number of Rs. 200 notes} = 4$$

$$\text{The number of Rs. 2000 notes} = 7 - 4 = 3$$

$$\text{The total amount deposited in a bank} = 5 * 20 + 4 * 200 + 3 * 2000 = \text{Rs. } 6900$$

Kavin:

$$\text{The sum of the number of Rs. 200 notes and the number of Rs. 2000 notes} = 24$$

$$\text{The ratio of the Rs. 20 notes to the Rs. 200 notes} = 3: 4 = 3b: 4b$$

$$\text{The number of Rs. 2000 notes} = 24 - 4b$$

Let,

$$20 * 3b + 4b * 200 + (24 - 4b) * 2000 = 19440$$

$$60b + 800b + 48000 - 8000b = 19440$$

$$7140b = 28560$$

$$b = 4$$

$$\text{The number of Rs. 20 notes} = 3b = 12$$

$$\text{The number of Rs. 200 notes} = 4b = 16$$

$$\text{The number of Rs. 2000 notes} = 24 - 16 = 8$$

Suriya:

$$\text{The total number of currency notes} = 27$$

$$\text{The ratio of the Rs. 20 notes to the Rs. 200 notes} = 3: 4 = 3c: 4c$$

$$\text{The number of Rs. 2000 notes} = 27 - 3c - 4c = 27 - 7c$$

We cannot find the further more data

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Veera:

The sum of the number of Rs. 200 notes and the number of Rs. 2000 notes = 35

The ratio of the Rs. 20 notes to the Rs. 200 notes = 1: 2 = d: 2d

The number of Rs. 2000 notes = 35 – 2d

Let

$$20 * d + 2d * 200 + (35 - 2d) * 2000 = 34200$$

$$20d + 400d + 70000 - 4000d = 34200$$

$$3580d = 35800$$

$$d = 10$$

The number of Rs. 20 notes = d = 10

The number of Rs. 200 notes = 2d = 20

The number of Rs. 2000 notes = 35 – 20 = 15

Persons	The total number of notes	The number of Rs. 20 notes	The number of Rs. 200 notes	The number of Rs. 2000 notes	The total amount deposited (Rs.)
Raj	12	5	4	3	6900
Kavin	36	12	16	8	19440
Suriya	27	-	-	-	-
Veera	45	10	20	15	34200

Answer: B

The number of Rs. 20 notes deposited by Mani = 125% of 12 = 15

The number of Rs. 200 notes deposited by Mani = 150% of 4 = 6

The number of Rs. 2000 notes deposited by Mani = p

Let,

$$20 * 15 + 200 * 6 + 2000 * p = 25500$$

$$2000p = 24000$$

$$p = 12$$

The total number of notes deposited by Mani = 15 + 6 + 12 = 33

The total number of notes deposited by Kavin = 36 (he deposited second highest number of notes)

6. Questions

From pie chart,

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$$4x + x^2 - 3 + 18 + 6x + 2x = 100$$

$$x^2 + 12x - 85 = 0$$

$$x^2 + 17x - 5x - 85 = 0$$

$$x(x + 17) - 5(x + 17) = 0$$

$$(x + 17)(x - 5) = 0$$

$$x = 5 \text{ and } -17$$

We consider the positive number,

So,

$$x = 5$$

The total number of employees in five companies together is 'N * 10x'

The value of 'N' is a third smallest two-digit prime number.

$$N = 17$$

$$\text{The total number of employees in five companies together} = 17 * 10 * 5 = 850$$

Company A:

$$\text{The total number of employees} = 4x\% \text{ of } 850 = 20\% \text{ of } 850 = 170$$

$$\text{The number of full-time employees} = 60\% \text{ of } 170 = 102$$

$$\text{The number of intern employees} = 170 - 102 = 68$$

$$\text{The ratio of full-time male employees to intern female employees} = 5: 1$$

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company B:

$$\text{The total number of employees} = (x^2 - 3)\% \text{ of } 850 = 22\% \text{ of } 850 = 187$$

$$\text{The number of full-time employees} = 72.72\% \text{ of } 187 = \frac{8}{11} * 187 = 136$$

$$\text{The number of intern employees} = 187 - 136 = 51$$

$$\text{The ratio of full-time male employees to intern female employees} = 85: 18$$

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company C:

$$\text{The total number of employees} = 18\% \text{ of } 850 = 153$$

$$\text{The number of full-time employees} = 66.66\% \text{ of } 153 = \left(\frac{2}{3}\right) * 153 = 102$$

$$\text{The number of intern employees} = 153 - 102 = 51$$

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The ratio of full-time male employees to intern female employees = 4: 1

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company D:

The total number of employees = 6x% of 850 = 30% of 850 = 255

The number of full-time employees = 40% of 255 = 102

The number of intern employees = 255 – 102 = 153

The ratio of full-time male employees to intern female employees = 12: 17

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company E:

The total number of employees = 2x% of 850 = 10% of 850 = 85

The number of full-time employees = 80% of 85 = 68

The number of intern employees = 85 – 68 = 17

The ratio of full-time male employees to intern female employees = 17: 2

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Let table,

Company	The total number of employees	The number of full-time employees	The number of intern employees
A	170	102	68
B	187	136	51
C	153	102	51
D	255	102	153
E	85	68	17

Answer: A

Let according to the question,

The ratio of full-time male employees to intern female employees = 5: 1 = 5a: a

The number of full-time female employees in company A = The number of intern female employees in company A = a

Let,

$$5a + a = 102$$

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$$6a = 102$$

$$a = 17$$

Then,

The number of full-time female employees in company A = The number of intern female employees in company A = 17

The number of full-time male employees in company A = $5a = 85$

The number of intern male employees in company A = $68 - 17 = 51$

The number of full-time female employees in company B = 51

The number of full-time male employees in company B = $136 - 51 = 85$

The ratio of full-time male employees to intern female employees = 85: 18

The number of intern female employees in company B = $85/85 * 18 = 18$

The number of intern male employees in company B = $51 - 18 = 33$

Thus, average = $(33 + 51)/2 = 42$

7. Questions

From pie chart,

$$4x + x^2 - 3 + 18 + 6x + 2x = 100$$

$$x^2 + 12x - 85 = 0$$

$$x^2 + 17x - 5x - 85 = 0$$

$$x(x + 17) - 5(x + 17) = 0$$

$$(x + 17)(x - 5) = 0$$

$$x = 5 \text{ and } -17$$

We consider the positive number,

So,

$$x = 5$$

@QuizByMickey

The total number of employees in five companies together is 'N * 10x'

The value of 'N' is a third smallest two-digit prime number.

$$N = 17$$

The total number of employees in five companies together = $17 * 10 * 5 = 850$

Company A:

The total number of employees = $4x\%$ of 850 = 20% of 850 = 170

The number of full-time employees = 60% of 170 = 102

The number of intern employees = $170 - 102 = 68$

The ratio of full-time male employees to intern female employees = 5: 1

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company B:

The total number of employees = $(x^2 - 3)\%$ of 850 = 22% of 850 = 187

The number of full-time employees = 72.72% of 187 = $\frac{8}{11} * 187 = 136$

The number of intern employees = $187 - 136 = 51$

The ratio of full-time male employees to intern female employees = 85: 18

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company C:

The total number of employees = 18% of 850 = 153

The number of full-time employees = 66.66% of 153 = $(\frac{2}{3}) * 153 = 102$

The number of intern employees = $153 - 102 = 51$

The ratio of full-time male employees to intern female employees = 4: 1

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company D:

The total number of employees = $6x\%$ of 850 = 30% of 850 = 255

The number of full-time employees = 40% of 255 = 102

The number of intern employees = $255 - 102 = 153$

The ratio of full-time male employees to intern female employees = 12: 17

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company E:

The total number of employees = $2x\%$ of 850 = 10% of 850 = 85

The number of full-time employees = 80% of 85 = 68

The number of intern employees = $85 - 68 = 17$

The ratio of full-time male employees to intern female employees = 17: 2

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

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Let table,

Company	The total number of employees	The number of full-time employees	The number of intern employees
A	170	102	68
B	187	136	51
C	153	102	51
D	255	102	153
E	85	68	17

Answer: C

The total number of full-time female employees in company B = 51

The number of full-time male employees = $136 - 51 = 85$

The number of intern female employees = $85/85 * 18 = 18$

The number of intern male employees = $51 - 18 = 33$

I: The number of intern male employees in company B is a multiple of 3

Let,

33 is a multiple of 3 (true)

II: The number of intern female employees in company B is multiple of 2

Let,

18 is a multiple of 2 (true)

III: The number of full-time male employees in company B is multiple of 5

Let,

85 is a multiple of 5 (true).

8. Questions

From pie chart,

$$4x + x^2 - 3 + 18 + 6x + 2x = 100$$

$$x^2 + 12x - 85 = 0$$

$$x^2 + 17x - 5x - 85 = 0$$

$$x(x + 17) - 5(x + 17) = 0$$

$$(x + 17)(x - 5) = 0$$

$$x = 5 \text{ and } -17$$

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We consider the positive number,

So,

$$x = 5$$

The total number of employees in five companies together is ' $N * 10x$ '

The value of 'N' is a third smallest two-digit prime number.

$$N = 17$$

The total number of employees in five companies together = $17 * 10 * 5 = 850$

Company A:

The total number of employees = $4x\%$ of 850 = 20% of 850 = 170

The number of full-time employees = 60% of 170 = 102

The number of intern employees = $170 - 102 = 68$

The ratio of full-time male employees to intern female employees = 5: 1

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company B:

The total number of employees = $(x^2 - 3)\%$ of 850 = 22% of 850 = 187

The number of full-time employees = 72.72% of 187 = $8/11 * 187 = 136$

The number of intern employees = $187 - 136 = 51$

The ratio of full-time male employees to intern female employees = 85: 18

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company C:

The total number of employees = 18% of 850 = 153

The number of full-time employees = 66.66% of 153 = $(2/3) * 153 = 102$

The number of intern employees = $153 - 102 = 51$

The ratio of full-time male employees to intern female employees = 4: 1

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company D:

The total number of employees = $6x\%$ of 850 = 30% of 850 = 255

The number of full-time employees = 40% of 255 = 102

The number of intern employees = $255 - 102 = 153$

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The ratio of full-time male employees to intern female employees = 12: 17

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company E:

The total number of employees = 2x% of 850 = 10% of 850 = 85

The number of full-time employees = 80% of 85 = 68

The number of intern employees = 85 – 68 = 17

The ratio of full-time male employees to intern female employees = 17: 2

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Let table,

Company	The total number of employees	The number of full-time employees	The number of intern employees
A	170	102	68
B	187	136	51
C	153	102	51
D	255	102	153
E	85	68	17

Answer: E

The sum of the fulltime employees in companies D and E together = 102 + 68 = 170

The sum of the intern employees in companies A, B and C together = 68 + 51 + 51 = 170

Thus percentage = (170/170) * 100 = 100%

9. Questions

From pie chart,

$$4x + x^2 - 3 + 18 + 6x + 2x = 100$$

$$x^2 + 12x - 85 = 0$$

$$x^2 + 17x - 5x - 85 = 0$$

$$x(x + 17) - 5(x + 17) = 0$$

$$(x + 17)(x - 5) = 0$$

$$x = 5 \text{ and } -17$$

We consider the positive number,

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So,

$$x = 5$$

The total number of employees in five companies together is ' $N * 10x$ '

The value of 'N' is a third smallest two-digit prime number.

$$N = 17$$

The total number of employees in five companies together = $17 * 10 * 5 = 850$

Company A:

The total number of employees = $4x\%$ of 850 = 20% of 850 = 170

The number of full-time employees = 60% of 170 = 102

The number of intern employees = $170 - 102 = 68$

The ratio of full-time male employees to intern female employees = 5: 1

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company B:

The total number of employees = $(x^2 - 3)\%$ of 850 = 22% of 850 = 187

The number of full-time employees = 72.72% of 187 = $8/11 * 187 = 136$

The number of intern employees = $187 - 136 = 51$

The ratio of full-time male employees to intern female employees = 85: 18

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company C:

The total number of employees = 18% of 850 = 153

The number of full-time employees = 66.66% of 153 = $(2/3) * 153 = 102$

The number of intern employees = $153 - 102 = 51$

The ratio of full-time male employees to intern female employees = 4: 1

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company D:

The total number of employees = $6x\%$ of 850 = 30% of 850 = 255

The number of full-time employees = 40% of 255 = 102

The number of intern employees = $255 - 102 = 153$

The ratio of full-time male employees to intern female employees = 12: 17

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company E:

The total number of employees = $2x\%$ of 850 = 10% of 850 = 85

The number of full-time employees = 80% of 85 = 68

The number of intern employees = $85 - 68 = 17$

The ratio of full-time male employees to intern female employees = 17: 2

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Let table,

Company	The total number of employees	The number of full-time employees	The number of intern employees
A	170	102	68
B	187	136	51
C	153	102	51
D	255	102	153
E	85	68	17

Answer: D

The ratio of full-time male employees to intern female employees = $4c: c$

The ratio of full-time male employees to intern female employees = $12d: 17d$

The sum of the number of full-time male employees in companies C and D together = $28x = 28 * 5 = 140$

The sum of the number of intern female employees in companies C and D together = $20 * 5 + 19 = 119$

Let,

$$4c + 12d = 140$$

$$c + 3d = 35 \text{ -----(1)}$$

and,

$$c + 17d = 119 \text{ ----- (2)}$$

$$(2) - (1)$$

$$17d - 3d = 119 - 35$$

$$14d = 84$$

$$d = 6$$

and

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$$c + 3 * 6 = 35$$

$$c = 17$$

The ratio of full-time male employees to intern female employees = 4c: c

The number of full-time male employees in company C = 4c = 68

The number of full-time female employees in company C = 102 – 68 = 34

The number of intern female employees in company C = c = 17

The number of intern male employees in company C = 51 – 17 = 34

The ratio of full-time male employees to intern female employees = 12d: 17d

The number of full-time male employees in company D = 12d = 72

The number of full-time female employees in company D = 102 – 72 = 30

The number of intern female employees in company D = 17d = 102

The number of intern male employees in company D = 153 – 102 = 51

I: The number of full-time male employees in company C > The number of full-time male employees in company D

$$68 > 72$$

This is not satisfied

II: The number of full-time female employees in company C > The number of intern male employees in company D

$$34 > 51$$

This is not satisfied

III: The number of intern female employees in company C < The number of full-time female employees in company D

$$17 < 30$$

This is satisfied.

10. Questions

From pie chart,

$$4x + x^2 - 3 + 18 + 6x + 2x = 100$$

$$x^2 + 12x - 85 = 0$$

$$x^2 + 17x - 5x - 85 = 0$$

$$x(x + 17) - 5(x + 17) = 0$$

$$(x + 17)(x - 5) = 0$$

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$$x = 5 \text{ and } -17$$

We consider the positive number,

So,

$$x = 5$$

The total number of employees in five companies together is ' $N * 10x$ '

The value of 'N' is a third smallest two-digit prime number.

$$N = 17$$

$$\text{The total number of employees in five companies together} = 17 * 10 * 5 = 850$$

Company A:

$$\text{The total number of employees} = 4x\% \text{ of } 850 = 20\% \text{ of } 850 = 170$$

$$\text{The number of full-time employees} = 60\% \text{ of } 170 = 102$$

$$\text{The number of intern employees} = 170 - 102 = 68$$

$$\text{The ratio of full-time male employees to intern female employees} = 5: 1$$

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company B:

$$\text{The total number of employees} = (x^2 - 3)\% \text{ of } 850 = 22\% \text{ of } 850 = 187$$

$$\text{The number of full-time employees} = 72.72\% \text{ of } 187 = \frac{8}{11} * 187 = 136$$

$$\text{The number of intern employees} = 187 - 136 = 51$$

$$\text{The ratio of full-time male employees to intern female employees} = 85: 18$$

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company C:

$$\text{The total number of employees} = 18\% \text{ of } 850 = 153$$

$$\text{The number of full-time employees} = 66.66\% \text{ of } 153 = \left(\frac{2}{3}\right) * 153 = 102$$

$$\text{The number of intern employees} = 153 - 102 = 51$$

$$\text{The ratio of full-time male employees to intern female employees} = 4: 1$$

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company D:

$$\text{The total number of employees} = 6x\% \text{ of } 850 = 30\% \text{ of } 850 = 255$$

$$\text{The number of full-time employees} = 40\% \text{ of } 255 = 102$$

The number of intern employees = $255 - 102 = 153$

The ratio of full-time male employees to intern female employees = 12: 17

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Company E:

The total number of employees = $2x\%$ of 850 = 10% of 850 = 85

The number of full-time employees = 80% of 85 = 68

The number of intern employees = $85 - 68 = 17$

The ratio of full-time male employees to intern female employees = 17: 2

We cannot find the number of male (full-time or intern) employees and the number of female (full-time or intern) employees.

Let table,

Company	The total number of employees	The number of full-time employees	The number of intern employees
A	170	102	68
B	187	136	51
C	153	102	51
D	255	102	153
E	85	68	17

Answer: C

The total number of employees in company F = 120% of 85 = 102 ($4x = 20$)

The number of intern employees in company F = $102/17 * 5 = 30$

The number of full-time female employees in company E = $30 + 4 = 34$

The number of full-time male employees in company E = $68 - 34 = 34$

The number of intern female employees in company E = $34/17 * 2 = 4$

The number of intern male employees in company E = $17 - 4 = 13$

Thus, ratio = 34: 13

11. Questions

The value of 'x' is a two-digit number and 'x' is a multiple of 10 and $10 < x < 40$

The possible values of 'x' is 20 or 30

Case (i): $x = 30$

The cost price of the product C = Rs. $90x$ = Rs. 2700

The marked price of product C is $[(3x + 5)\% = 95\%]$ more than its cost price and the discount percentage of the product C is 30%.

The marked price of the product C = 195% of 2700 = Rs. 5265

The selling price of the product C = 70% of 5265 = Rs. 3685.5

This is not satisfied.

Case (ii): $x = 20$

The cost price of the product C = Rs. $90x$ = Rs. 1800

The marked price of product C is $[(3x + 5)\% = 65\%]$ more than its cost price and the discount percentage of the product C is 20%.

The marked price of the product C = 165% of 1800 = Rs. 2970

The selling price of the product C = 80% of 2970 = Rs. 2376

This is an integer. So, $x = 20$

The profit amount of product C is Rs. C^2

$$C^2 = 2376 - 1800 = 576$$

$$C^2 = 576$$

$$C = 24$$

Let,

The ratio of the selling price of products B and C is 49:36

The selling price of product B = $2376/36 * 49$ = Rs. 3234

The ratio of the discount percentage of products B and C is 4: 5

The discount percentage of product C = $20/5 * 4$ = 16%

The marked price of the product B = $3234/84 * 100$ = Rs. 3850

The cost price of product B which marked up 54% above its cost price

The cost price of the product B = $3850/154 * 100$ = Rs. 2500

Let,

The marked up price of the product A = Rs. $62x$ = $62 * 20$ = Rs. 1240

Let assume the cost price of the product A = Rs. $100a$

The profit percentage of product A is 37.7% and the discount percentage of the product A is $[(x - 5)\% = 15\%]$

The selling price of the product A = 137.7% of $100a$ = Rs. $137.7a$

The marked price of the product A = $(137.7a/85) * 100$ = Rs. $162a$

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The marked up price of the product A = $162a - 100a = \text{Rs. } 62a$

Then,

$$62a = 1240$$

$$a = 20$$

Then,

The cost price of the product A = $\text{Rs. } 100a = \text{Rs. } 2000$

The marked price of the product A = $162 * 20 = \text{Rs. } 3240$

The selling price of the product A = $137.7 * 20 = \text{Rs. } 2754$

Then,

The ratio of the discount percentage of products A and C is B: D respectively

$$B: D = 15: 20 = 3: 4$$

Then,

The cost price of product A is A% less than the cost price of product B.

$$A = [(2500 - 2000)/2500] * 100$$

$$A = 20$$

The difference between the cost price of products A and C is E.

$$E = 2000 - 1800$$

$$E = 200$$

Let table,

Product	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)
A	2000	3240	2754
B	2500	3850	3234
C	1800	2970	2376

Answer: A

@QuizByMickey

Let according to the question,

I: The selling price of the product B is a multiple of D

Let,

$$3234/4 = 808.5$$

This is not satisfied.

II: The value of (A + B) is a prime number.

$$20 + 3 = 23 \text{ (prime number)}$$

This is satisfied.

III: The value of 'x' is a multiple of B.

20 is not a multiple of 3.

This is not satisfied.

12. Questions

The value of 'x' is a two-digit number and 'x' is a multiple of 10 and $10 < x < 40$

The possible values of 'x' is 20 or 30

Case (i): $x = 30$

The cost price of the product C = Rs. $90x = \text{Rs. } 2700$

The marked price of product C is $[(3x + 5)\% = 95\%]$ more than its cost price and the discount percentage of the product C is 30%.

The marked price of the product C = 195% of 2700 = Rs. 5265

The selling price of the product C = 70% of 5265 = Rs. 3685.5

This is not satisfied.

Case (ii): $x = 20$

The cost price of the product C = Rs. $90x = \text{Rs. } 1800$

The marked price of product C is $[(3x + 5)\% = 65\%]$ more than its cost price and the discount percentage of the product C is 20%.

The marked price of the product C = 165% of 1800 = Rs. 2970

The selling price of the product C = 80% of 2970 = Rs. 2376

This is an integer. So, $x = 20$

The profit amount of product C is Rs. C^2

$$C^2 = 2376 - 1800 = 576$$

$$C^2 = 576$$

$$C = 24$$

Let,

The ratio of the selling price of products B and C is 49:36

The selling price of product B = $2376/36 * 49 = \text{Rs. } 3234$

The ratio of the discount percentage of products B and C is 4: 5

The discount percentage of product C = $20/5 * 4 = 16\%$

The marked price of the product B = $3234/84 * 100 = \text{Rs. } 3850$

@QuizByMickey

The cost price of product B which marked up 54% above its cost price

The cost price of the product B = $3850/154 * 100 = \text{Rs. } 2500$

Let,

The marked up price of the product A = Rs. $62x = 62 * 20 = \text{Rs. } 1240$

Let assume the cost price of the product A = Rs. $100a$

The profit percentage of product A is 37.7% and the discount percentage of the product A is $[(x - 5)\% = 15\%]$

The selling price of the product A = 137.7% of $100a = \text{Rs. } 137.7a$

The marked price of the product A = $(137.7a/85) * 100 = \text{Rs. } 162a$

The marked up price of the product A = $162a - 100a = \text{Rs. } 62a$

Then,

$$62a = 1240$$

$$a = 20$$

Then,

The cost price of the product A = Rs. $100a = \text{Rs. } 2000$

The marked price of the product A = $162 * 20 = \text{Rs. } 3240$

The selling price of the product A = $137.7 * 20 = \text{Rs. } 2754$

Then,

The ratio of the discount percentage of products A and C is __B: D__ respectively

$$B: D = 15: 20 = 3: 4$$

Then,

The cost price of product A is __A__% less than the cost price of product B.

$$A = [(2500 - 2000)/2500] * 100$$

$$A = 20$$

The difference between the cost price of products A and C is Rs. __E__.

$$E = 2000 - 1800$$

$$E = 200$$

Let table,

Product	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)
A	2000	3240	2754
B	2500	3850	3234
C	1800	2970	2376

Answer: E

New discount = $(A + 5)\% = (20 + 5)\% = 25\%$

The new selling price of the product C = 75% of 2376 = Rs. 1782

The new loss percentage = $[(1800 - 1782)/1800] * 100 = 1\%$

$B - 2 = 3 - 2 = 1$

13. Questions

The value of 'x' is a two-digit number and 'x' is a multiple of 10 and $10 < x < 40$

The possible values of 'x' is 20 or 30

Case (i): $x = 30$

The cost price of the product C = Rs. $90x = \text{Rs. } 2700$

The marked price of product C is $[(3x + 5)\% = 95\%]$ more than its cost price and the discount percentage of the product C is 30%.

The marked price of the product C = 195% of 2700 = Rs. 5265

The selling price of the product C = 70% of 5265 = Rs. 3685.5

This is not satisfied.

Case (ii): $x = 20$

The cost price of the product C = Rs. $90x = \text{Rs. } 1800$

The marked price of product C is $[(3x + 5)\% = 65\%]$ more than its cost price and the discount percentage of the product C is 20%.

The marked price of the product C = 165% of 1800 = Rs. 2970

The selling price of the product C = 80% of 2970 = Rs. 2376 @QuizByMickey

This is an integer. So, $x = 20$

The profit amount of product C is Rs. C^2

$$C^2 = 2376 - 1800 = 576$$

$$C^2 = 576$$

$$C = 24$$

Let,

The ratio of the selling price of products B and C is 49:36

The selling price of product B = $2376/36 * 49 = \text{Rs. } 3234$

The ratio of the discount percentage of products B and C is 4: 5

The discount percentage of product C = $20/5 * 4 = 16\%$

The marked price of the product B = $3234/84 * 100 = \text{Rs. } 3850$

The cost price of product B which marked up 54% above its cost price

The cost price of the product B = $3850/154 * 100 = \text{Rs. } 2500$

Let,

The marked up price of the product A = Rs. $62x = 62 * 20 = \text{Rs. } 1240$

Let assume the cost price of the product A = Rs. 100a

The profit percentage of product A is 37.7% and the discount percentage of the product A is $[(x - 5)\% = 15\%]$

The selling price of the product A = 137.7% of 100a = Rs. 137.7a

The marked price of the product A = $(137.7a/85) * 100 = \text{Rs. } 162a$

The marked up price of the product A = $162a - 100a = \text{Rs. } 62a$

Then,

$$62a = 1240$$

$$a = 20$$

Then,

The cost price of the product A = Rs. 100a = Rs. 2000

The marked price of the product A = $162 * 20 = \text{Rs. } 3240$

The selling price of the product A = $137.7 * 20 = \text{Rs. } 2754$

Then,

The ratio of the discount percentage of products A and C is __B: D__ respectively

$$B: D = 15: 20 = 3: 4$$

Then,

The cost price of product A is __A__% less than the cost price of product B.

$$A = [(2500 - 2000)/2500] * 100$$

$$A = 20$$

The difference between the cost price of products A and C is Rs. __E__.

$$E = 2000 - 1800$$

@QuizByMickey

$E = 200$

Let table,

Product	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)
A	2000	3240	2754
B	2500	3850	3234
C	1800	2970	2376

Answer: D

The difference between the selling price of the products A and B = $3234 - 2754 = \text{Rs. } 480$

$F = 480$

Let,

Series: 4, 20, 24, $(200 - 75)$, 130, $(480 + 170)$

4, 20, 24, 125, 130, 650

4	20	25	125	130	650
	*5	+5	*5	+5	*5

Thus, wrong number is $C = 24$.

14. Questions

The value of 'x' is a two-digit number and 'x' is a multiple of 10 and $10 < x < 40$

The possible values of 'x' is 20 or 30

Case (i): $x = 30$

The cost price of the product C = Rs. $90x = \text{Rs. } 2700$

The marked price of product C is $[(3x + 5)\% = 95\%]$ more than its cost price and the discount percentage of the product C is 30%.

The marked price of the product C = 195% of 2700 = Rs. 5265

The selling price of the product C = 70% of 5265 = Rs. 3685.5

This is not satisfied.

Case (ii): $x = 20$

The cost price of the product C = Rs. $90x = \text{Rs. } 1800$

The marked price of product C is $[(3x + 5)\% = 65\%]$ more than its cost price and the discount percentage of the product C is 20%.

The marked price of the product C = 165% of 1800 = Rs. 2970

The selling price of the product C = 80% of 2970 = Rs. 2376

@QuizByMickey

This is an integer. So, $x = 20$

The profit amount of product C is Rs. C^2

$$C^2 = 2376 - 1800 = 576$$

$$C^2 = 576$$

$$C = 24$$

Let,

The ratio of the selling price of products B and C is 49:36

The selling price of product B = $2376/36 * 49 = \text{Rs. } 3234$

The ratio of the discount percentage of products B and C is 4: 5

The discount percentage of product C = $20/5 * 4 = 16\%$

The marked price of the product B = $3234/84 * 100 = \text{Rs. } 3850$

The cost price of product B which marked up 54% above its cost price

The cost price of the product B = $3850/154 * 100 = \text{Rs. } 2500$

Let,

The marked up price of the product A = Rs. $62x = 62 * 20 = \text{Rs. } 1240$

Let assume the cost price of the product A = Rs. $100a$

The profit percentage of product A is 37.7% and the discount percentage of the product A is $[(x - 5)\% = 15\%]$

The selling price of the product A = 137.7% of $100a = \text{Rs. } 137.7a$

The marked price of the product A = $(137.7a/85) * 100 = \text{Rs. } 162a$

The marked up price of the product A = $162a - 100a = \text{Rs. } 62a$

Then,

$$62a = 1240$$

$$a = 20$$

Then,

The cost price of the product A = Rs. $100a = \text{Rs. } 2000$

The marked price of the product A = $162 * 20 = \text{Rs. } 3240$

The selling price of the product A = $137.7 * 20 = \text{Rs. } 2754$

Then,

The ratio of the discount percentage of products A and C is __B: D__ respectively

$$B: D = 15: 20 = 3: 4$$

@QuizByMickey

Then,

The cost price of product A is __A__% less than the cost price of product B.

$$A = [(2500 - 2000)/2500] * 100$$

$$A = 20$$

The difference between the cost price of products A and C is Rs. __E__.

$$E = 2000 - 1800$$

$$E = 200$$

Let table,

Product	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)
A	2000	3240	2754
B	2500	3850	3234
C	1800	2970	2376

Answer: C

The cost price of the product E = 120% of 1800 = Rs. 2160

The marked price of the product E = 150% of 2160 = Rs. 3240

$$A = 20 \text{ and } C + 1 = 25$$

The selling price of the product E = $3240 * 80/100 * 75/100$ = Rs. 1944

The profit or loss percentage = $[(1944 - 2160)/2160] * 100 = 10\%$

15. Questions

The value of 'x' is a two-digit number and 'x' is a multiple of 10 and $10 < x < 40$

The possible values of 'x' is 20 or 30

Case (i): $x = 30$

The cost price of the product C = Rs. $90x$ = Rs. 2700

The marked price of product C is $[(3x + 5)\% = 95\%]$ more than its cost price and the discount percentage of the product C is 30%.

The marked price of the product C = 195% of 2700 = Rs. 5265

The selling price of the product C = 70% of 5265 = Rs. 3685.5

This is not satisfied.

Case (ii): $x = 20$

The cost price of the product C = Rs. $90x$ = Rs. 1800

The marked price of product C is $[(3x + 5)\% = 65\%]$ more than its cost price and the discount percentage

of the product C is 20%.

The marked price of the product C = 165% of 1800 = Rs. 2970

The selling price of the product C = 80% of 2970 = Rs. 2376

This is an integer. So, $x = 20$

The profit amount of product C is Rs. C^2

$$C^2 = 2376 - 1800 = 576$$

$$C^2 = 576$$

$$C = 24$$

Let,

The ratio of the selling price of products B and C is 49:36

The selling price of product B = $2376/36 * 49$ = Rs. 3234

The ratio of the discount percentage of products B and C is 4: 5

The discount percentage of product C = $20/5 * 4$ = 16%

The marked price of the product B = $3234/84 * 100$ = Rs. 3850

The cost price of product B which marked up 54% above its cost price

The cost price of the product B = $3850/154 * 100$ = Rs. 2500

Let,

The marked up price of the product A = Rs. $62x = 62 * 20$ = Rs. 1240

Let assume the cost price of the product A = Rs. 100a

The profit percentage of product A is 37.7% and the discount percentage of the product A is $[(x - 5)\% = 15\%]$

The selling price of the product A = 137.7% of 100a = Rs. 137.7a

The marked price of the product A = $(137.7a/85) * 100$ = Rs. 162a

The marked up price of the product A = $162a - 100a$ = Rs. 62a @QuizByMickey

Then,

$$62a = 1240$$

$$a = 20$$

Then,

The cost price of the product A = Rs. 100a = Rs. 2000

The marked price of the product A = $162 * 20$ = Rs. 3240

The selling price of the product A = $137.7 * 20$ = Rs. 2754

Then,

The ratio of the discount percentage of products A and C is ___B: D___ respectively

$$B: D = 15: 20 = 3: 4$$

Then,

The cost price of product A is ___A___% less than the cost price of product B.

$$A = [(2500 - 2000)/2500] * 100$$

$$A = 20$$

The difference between the cost price of products A and C is Rs. ___E___.

$$E = 2000 - 1800$$

$$E = 200$$

Let table,

Product	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)
A	2000	3240	2754
B	2500	3850	3234
C	1800	2970	2376

Answer: E

The selling of the product D = 24% of $(3240 + 3850 + 2970)/3 = \text{Rs. } 804.8$

The cost price of the product D = $804.8/80 * 100 = \text{Rs. } 1006$

16. Questions

In screen 1:

The number of seats = $2a$

The number of rows = $b + 5$

The number of seats in each row in screen 1 = 8

Let

$$(b + 5) * 8 = 2a$$

$$8b + 40 = 2a$$

$$2a - 8b = 40$$

$$a - 4b = 20 \text{ ----- (1)}$$

In screen 4

The number of seats = $3a$

The number of rows = $b - 5$

@QuizByMickey

The number of seats in each row in screen 1 = 20

Let

$$(b - 5) * 20 = 3a$$

$$20b - 100 = 3a$$

$$20b - 3a = 100 \text{ ----- (2)}$$

Let

$$(1) * 3 \rightarrow 3a - 12b = 60$$

Then, solve (1) and (2)

$$20b - 12b = 100 + 60$$

$$8b = 160$$

$$b = 20$$

and,

From (1)

$$a - 4 * 20 = 20$$

$$a = 100$$

Then,

In screen 1:

$$\text{The number of seats} = 2a = 200$$

$$\text{The number of rows} = b + 5 = 25$$

$$\text{The number of seats in each row in screen 1} = 8$$

$$\text{The occupancy percentage (one show)} = (a-10)\% = 90\%$$

$$\text{The number of booked seats} = 90\% \text{ of } 200 = 180$$

$$\text{The number of not booked seats} = 200 - 180 = 20$$

In screen 2:

$$\text{The number of seats} = 3a - 20 = 300 - 20 = 280$$

$$\text{The number of rows} = b = 20$$

$$\text{The number of seats in each row in screen 1} = 280/20 = 14$$

$$\text{The occupancy percentage (one show)} = 85\%$$

$$\text{The number of booked seats} = 85\% \text{ of } 280 = 238$$

$$\text{The number of not booked seats} = 280 - 238 = 42$$

In screen 3:

@QuizByMickey

The number of seats = $2a + 20 = 200 + 20 = 220$

The number of rows = $b = 22$

The number of seats in each row in screen 1 = $220/22 = 10$

The occupancy percentage (one show) = $(a - b - 5)\% = (100 - 20 - 5) = 75\%$

The number of booked seats = 75% of $220 = 165$

The number of not booked seats = $220 - 165 = 55$

In screen 4

The number of seats = $3a = 300$

The number of rows = $b - 5 = 15$

The number of seats in each row in screen 1 = $300/15 = 20$

The occupancy percentage (one show) = 92%

The number of booked seats = 92% of $300 = 276$

The number of not booked seats = $300 - 276 = 24$

Let table,

Screen	The number of seats	The number of booked seats	The number of not booked seats	The number of rows	The number of seats in each row
1	200	180	20	25	8
2	280	238	42	20	14
3	220	165	55	22	10
4	300	276	24	15	20

Answer: B

Let according to the question,

The number of executive seats rows = 20% of $25 = 5$

The number of executive seats = $5 * 8 = 40$

The number of normal seats rows = $25 - 5 = 20$

The number of normal seats = $20 * 8 = 160$

The number of booked tickets in normal seats = $160 - 20 = 140$

Thus, total revenue = $40 * 350 + 140 * 200 = 14000 + 28000 = \text{Rs. } 42000$

17. Questions

In screen 1:

@QuizByMickey

The number of seats = $2a$

The number of rows = $b + 5$

The number of seats in each row in screen 1 = 8

Let

$$(b + 5) * 8 = 2a$$

$$8b + 40 = 2a$$

$$2a - 8b = 40$$

$$a - 4b = 20 \text{ ----- (1)}$$

In screen 4

The number of seats = $3a$

The number of rows = $b - 5$

The number of seats in each row in screen 1 = 20

Let

$$(b - 5) * 20 = 3a$$

$$20b - 100 = 3a$$

$$20b - 3a = 100 \text{ ----- (2)}$$

Let

$$(1) * 3 \rightarrow 3a - 12b = 60$$

Then, solve (1) and (2)

$$20b - 12b = 100 + 60$$

$$8b = 160$$

$$b = 20$$

and,

From (1)

$$a - 4 * 20 = 20$$

$$a = 100$$

Then,

In screen 1:

The number of seats = $2a = 200$

The number of rows = $b + 5 = 25$

The number of seats in each row in screen 1 = 8

@QuizByMickey

The occupancy percentage (one show) = $(a-10)\% = 90\%$

The number of booked seats = 90% of $200 = 180$

The number of not booked seats = $200 - 180 = 20$

In screen 2:

The number of seats = $3a - 20 = 300 - 20 = 280$

The number of rows = $b = 20$

The number of seats in each row in screen 1 = $280/20 = 14$

The occupancy percentage (one show) = 85%

The number of booked seats = 85% of $280 = 238$

The number of not booked seats = $280 - 238 = 42$

In screen 3:

The number of seats = $2a + 20 = 200 + 20 = 220$

The number of rows = $b = 22$

The number of seats in each row in screen 1 = $220/22 = 10$

The occupancy percentage (one show) = $(a - b - 5)\% = (100 - 20 - 5) = 75\%$

The number of booked seats = 75% of $220 = 165$

The number of not booked seats = $220 - 165 = 55$

In screen 4

The number of seats = $3a = 300$

The number of rows = $b - 5 = 15$

The number of seats in each row in screen 1 = $300/15 = 20$

The occupancy percentage (one show) = 92%

The number of booked seats = 92% of $300 = 276$

The number of not booked seats = $300 - 276 = 24$

Let table,

@QuizByMickey

Screen	The number of seats	The number of booked seats	The number of not booked seats	The number of rows	The number of seats in each row
1	200	180	20	25	8
2	280	238	42	20	14
3	220	165	55	22	10
4	300	276	24	15	20

Answer: E

The total number of seats booked in screen 2 by online = 57.14% of 238 = $(4/7) * 238 = 136$

The total number of seats booked in screen 2 by offline = $238 - 136 = 102$

Thus, percentage = $(102/300) * 100 = 34\%$

a). $(a - 66)\% = (100 - 66) = 34\%$

b). 34% of a = 34% of 100 = 34%

c). $(b + 14)\% = (20 + 14)\% = 34\%$

18. Questions

In screen 1:

The number of seats = $2a$

The number of rows = $b + 5$

The number of seats in each row in screen 1 = 8

Let

$$(b + 5) * 8 = 2a$$

$$8b + 40 = 2a$$

$$2a - 8b = 40$$

$$a - 4b = 20 \text{ ----- (1)}$$

In screen 4

The number of seats = $3a$

The number of rows = $b - 5$

The number of seats in each row in screen 1 = 20

Let

$$(b - 5) * 20 = 3a$$

$$20b - 100 = 3a$$

@QuizByMickey

$$20b - 3a = 100 \text{ ----- (2)}$$

Let

$$(1) * 3 \rightarrow 3a - 12b = 60$$

Then, solve (1) and (2)

$$20b - 12b = 100 + 60$$

$$8b = 160$$

$$b = 20$$

and,

From (1)

$$a - 4 * 20 = 20$$

$$a = 100$$

Then,

In screen 1:

$$\text{The number of seats} = 2a = 200$$

$$\text{The number of rows} = b + 5 = 25$$

$$\text{The number of seats in each row in screen 1} = 8$$

$$\text{The occupancy percentage (one show)} = (a-10)\% = 90\%$$

$$\text{The number of booked seats} = 90\% \text{ of } 200 = 180$$

$$\text{The number of not booked seats} = 200 - 180 = 20$$

In screen 2:

$$\text{The number of seats} = 3a - 20 = 300 - 20 = 280$$

$$\text{The number of rows} = b = 20$$

$$\text{The number of seats in each row in screen 1} = 280/20 = 14$$

$$\text{The occupancy percentage (one show)} = 85\%$$

$$\text{The number of booked seats} = 85\% \text{ of } 280 = 238$$

$$\text{The number of not booked seats} = 280 - 238 = 42$$

In screen 3:

$$\text{The number of seats} = 2a + 20 = 200 + 20 = 220$$

$$\text{The number of rows} = b = 22$$

$$\text{The number of seats in each row in screen 1} = 220/22 = 10$$

$$\text{The occupancy percentage (one show)} = (a - b - 5)\% = (100 - 20 - 5) = 75\%$$

@QuizByMickey

The number of booked seats = 75% of 220 = 165

The number of not booked seats = 220 – 165 = 55

In screen 4

The number of seats = 3a = 300

The number of rows = b - 5 = 15

The number of seats in each row in screen 1 = 300/15 = 20

The occupancy percentage (one show) = 92%

The number of booked seats = 92% of 300 = 276

The number of not booked seats = 300 – 276 = 24

Let table,

Screen	The number of seats	The number of booked seats	The number of not booked seats	The number of rows	The number of seats in each row
1	200	180	20	25	8
2	280	238	42	20	14
3	220	165	55	22	10
4	300	276	24	15	20

Answer: B

I: The number of rows in screen 3 is 22% of the value of 'a'.

Let,

22% of 100 = 22

This is true.

II: The number of not booked seats in screen 1 is 'b'.

20 = b

This is true.

III: The number of booked seats in screen 4 is '2a + 76'.

Let,

2a + 76 = 200 + 76 = 276

This is true.

19. Questions

In screen 1:

@QuizByMickey

The number of seats = $2a$

The number of rows = $b + 5$

The number of seats in each row in screen 1 = 8

Let

$$(b + 5) * 8 = 2a$$

$$8b + 40 = 2a$$

$$2a - 8b = 40$$

$$a - 4b = 20 \text{ ----- (1)}$$

In screen 4

The number of seats = $3a$

The number of rows = $b - 5$

The number of seats in each row in screen 1 = 20

Let

$$(b - 5) * 20 = 3a$$

$$20b - 100 = 3a$$

$$20b - 3a = 100 \text{ ----- (2)}$$

Let

$$(1) * 3 \rightarrow 3a - 12b = 60$$

Then, solve (1) and (2)

$$20b - 12b = 100 + 60$$

$$8b = 160$$

$$b = 20$$

and,

From (1)

$$a - 4 * 20 = 20$$

$$a = 100$$

Then,

In screen 1:

The number of seats = $2a = 200$

The number of rows = $b + 5 = 25$

The number of seats in each row in screen 1 = 8

@QuizByMickey

The occupancy percentage (one show) = $(a-10)\% = 90\%$

The number of booked seats = 90% of $200 = 180$

The number of not booked seats = $200 - 180 = 20$

In screen 2:

The number of seats = $3a - 20 = 300 - 20 = 280$

The number of rows = $b = 20$

The number of seats in each row in screen 1 = $280/20 = 14$

The occupancy percentage (one show) = 85%

The number of booked seats = 85% of $280 = 238$

The number of not booked seats = $280 - 238 = 42$

In screen 3:

The number of seats = $2a + 20 = 200 + 20 = 220$

The number of rows = $b = 22$

The number of seats in each row in screen 1 = $220/22 = 10$

The occupancy percentage (one show) = $(a - b - 5)\% = (100 - 20 - 5) = 75\%$

The number of booked seats = 75% of $220 = 165$

The number of not booked seats = $220 - 165 = 55$

In screen 4

The number of seats = $3a = 300$

The number of rows = $b - 5 = 15$

The number of seats in each row in screen 1 = $300/15 = 20$

The occupancy percentage (one show) = 92%

The number of booked seats = 92% of $300 = 276$

The number of not booked seats = $300 - 276 = 24$

Let table,

@QuizByMickey

Screen	The number of seats	The number of booked seats	The number of not booked seats	The number of rows	The number of seats in each row
1	200	180	20	25	8
2	280	238	42	20	14
3	220	165	55	22	10
4	300	276	24	15	20

Answer: C

The total number of seats in screen 5 = 75% of 300 = 225

The number of rows = the number of seats in each row in screen 5 = x

Let,

$$x * x = 225$$

$$x^2 = 225$$

$$x = 15$$

The number of rows in screen 5 = 15

Thus, required total = 22 + 15 + 15 = 52

20. Questions

In screen 1:

The number of seats = 2a

The number of rows = b + 5

The number of seats in each row in screen 1 = 8

Let

$$(b + 5) * 8 = 2a$$

$$8b + 40 = 2a$$

$$2a - 8b = 40$$

$$a - 4b = 20 \text{ ----- (1)}$$

In screen 4

The number of seats = 3a

The number of rows = b - 5

The number of seats in each row in screen 1 = 20

Let

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$$(b - 5) * 20 = 3a$$

$$20b - 100 = 3a$$

$$20b - 3a = 100 \text{ ----- (2)}$$

Let

$$(1) * 3 \rightarrow 3a - 12b = 60$$

Then, solve (1) and (2)

$$20b - 12b = 100 + 60$$

$$8b = 160$$

$$b = 20$$

and,

From (1)

$$a - 4 * 20 = 20$$

$$a = 100$$

Then,

In screen 1:

$$\text{The number of seats} = 2a = 200$$

$$\text{The number of rows} = b + 5 = 25$$

$$\text{The number of seats in each row in screen 1} = 8$$

$$\text{The occupancy percentage (one show)} = (a-10)\% = 90\%$$

$$\text{The number of booked seats} = 90\% \text{ of } 200 = 180$$

$$\text{The number of not booked seats} = 200 - 180 = 20$$

In screen 2:

$$\text{The number of seats} = 3a - 20 = 300 - 20 = 280$$

$$\text{The number of rows} = b = 20$$

$$\text{The number of seats in each row in screen 1} = 280/20 = 14$$

$$\text{The occupancy percentage (one show)} = 85\%$$

$$\text{The number of booked seats} = 85\% \text{ of } 280 = 238$$

$$\text{The number of not booked seats} = 280 - 238 = 42$$

In screen 3:

$$\text{The number of seats} = 2a + 20 = 200 + 20 = 220$$

$$\text{The number of rows} = b = 22$$

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The number of seats in each row in screen 1 = $220/22 = 10$

The occupancy percentage (one show) = $(a - b - 5)\% = (100 - 20 - 5) = 75\%$

The number of booked seats = 75% of $220 = 165$

The number of not booked seats = $220 - 165 = 55$

In screen 4

The number of seats = $3a = 300$

The number of rows = $b - 5 = 15$

The number of seats in each row in screen 1 = $300/15 = 20$

The occupancy percentage (one show) = 92%

The number of booked seats = 92% of $300 = 276$

The number of not booked seats = $300 - 276 = 24$

Let table,

Screen	The number of seats	The number of booked seats	The number of not booked seats	The number of rows	The number of seats in each row
1	200	180	20	25	8
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Answer: E

The average number of not booked seats in screens 2 and 4 in a particular show = $(42 + 24)/2 = 33$

The total number of seats in each row in screens 1, 2, 3, and 4 together = $8 + 14 + 10 + 20 = 52$

Thus, ratio = $33: 52$

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